Coursework Portfolio 1 — Week 5 Lab Report

Vulnerability Assessment: Introduction to NMAP and SSH(Secure Shell)

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Introduction

The aim of this lab was to understand and apply secure remote access using SSH and perform network reconnaissance using Nmap to identify hosts, open ports, services, and potential security vulnerabilities.

Introduction to SSH

SSH is a secure way to remotely access and control another computer over a network. It answers: How can I safely log into and control another machine over the internet?

1. Checking SSH Installation

The first step was verifying that SSH was installed on the system. This was done using the command: ssh -V

The output confirmed that the OpenSSH client was installed and available for use. SSH is a secure protocol that allows encrypted communication between systems over a network.

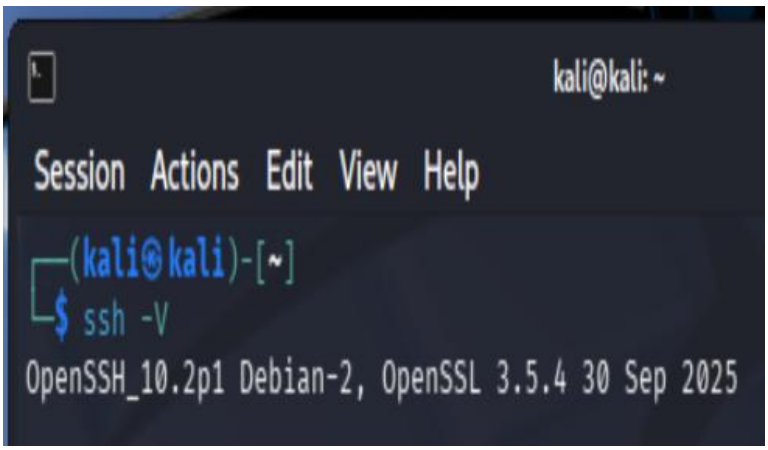


Figure 1:Verifying OpenSSH version installed on Kali Linux.

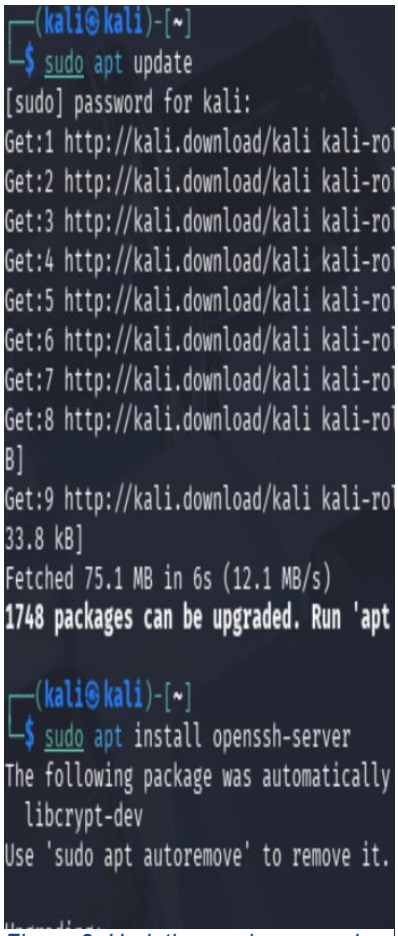


Figure 2: Updating packages and installing OpenSSH server.

2. Install SSH Server

To enable remote connections, the OpenSSH server package was installed using the following command:

sudo apt install openssh-server

This installs the SSH service that allows other machines to securely connect to the system.

3. Start SSH

After installing the SSH server, the service was started using:

sudo systemctl start ssh

The output confirmed that the SSH service was active and running.

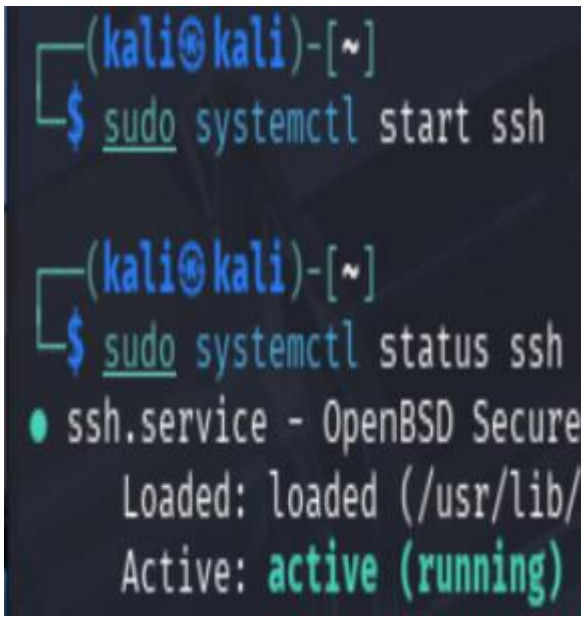


Figure 3: SSH service started and confirmed as active and running.

Figure 4: Kali Linux IP address identified as 192.168.123.102.

```
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
n 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel s
ault qlen 1000
    link/ether 08:00:27:76:05:9f brd ff:ff:ff:ff:ff:ff
    inet 192.168.123.102/24 brd 192.168.123.255 scope global dynamic
h0
       valid_lft 568sec preferred_lft 568sec
    inet6 fe80::3b02:c8b2:c1b3:36b3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel s
ault qlen 1000
    link/ether 08:00:27:b8:a1:0b brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixro
       valid_lft 85768sec preferred_lft 85768sec
    inet6 fe80::6574:3e39:28f2:129b/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

4. Check IP Address

To connect using SSH, the IP address of the machine was identified using the command:

ip a

The IP address is required so that another system can connect to the SSH service over the network.

In the screenshot we can see that the ip address is: 192.168.123.102

5. Connect to SSH

A secure connection was established using the SSH command:

ssh kali@localhost

This command connects to the local machine using the SSH protocol. After authentication with the system password, a secure session was created allowing commands to be executed remotely.

```
┌──(kali㉿kali)-[~]
└─$ ssh kali@localhost
kali@localhost's password:
Permission denied, please try again.
kali@localhost's password:
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24)
86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Mar 11 10:31:15 2026 from ::1
```

Figure 5: SSH connection to localhost successfully established after authentication.

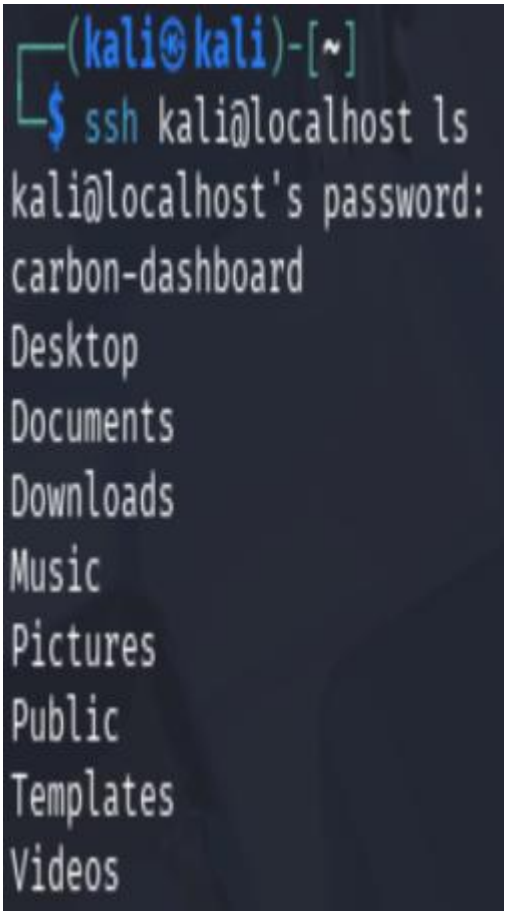


Figure 6: : Remote directory listing via SSH confirming successful access.

6. Execute command via SSH

SSH can also execute commands remotely without opening a full session. This was tested using:

ssh kali@localhost ls

The command connected to the system and executed the `ls` command to display the files in the directory.

7. Generate SSH key pair

To improve authentication security, an SSH key pair was generated using:

ssh-keygen -t rsa -b 4096

This created a public and private key used for passwordless authentication between systems.

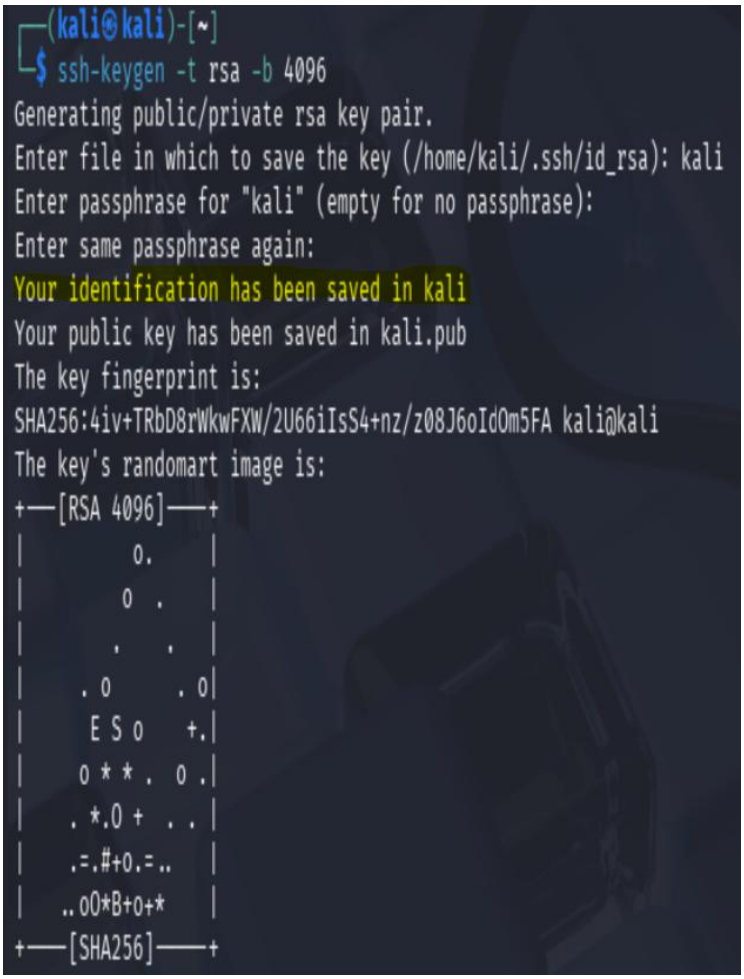


Figure 7: RSA 4096-bit key pair successfully generated.

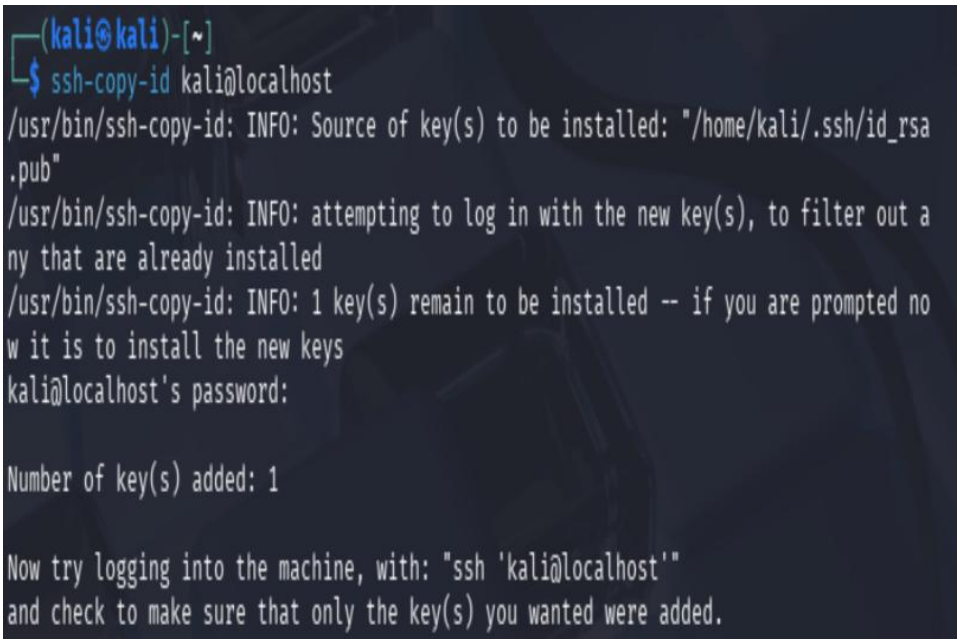


Figure 8: Public key copied to localhost using ssh-copy-id.

8. Copying the public key (SSH key)

The public key was copied to the SSH server using:

ssh-copy-id kali@localhost

This allows passwordless authentication by verifying the private key stored on the client machine.

9. Test Key-Based Authentication (Login)

After installing the public key, SSH login was tested again:

ssh kali@localhost

The connection was successful without requiring a password, demonstrating secure key-based authentication.

```
┌──(kali㉿kali)-[~]
└─$ ssh kali@localhost
Enter passphrase for key '/home/kali/.ssh/id_rsa':
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x
86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Mar 11 10:32:04 2026 from ::1
```

Figure 9: Passwordless SSH login achieved using the RSA key pair.

10. Secure file transfer

Files can be securely transferred using SCP. This was demonstrated using:

scp testfile.txt kali@localhost:/home/kali

```
┌──(kali㉿kali)-[~]
└─$ touch testfile.txt

┌──(kali㉿kali)-[~]
└─$ scp testfile.txt kali@localhost:/home/kali
Enter passphrase for key '/home/kali/.ssh/id_rsa':
testfile.txt                                  100%    0     0.0KB/s   00:00
```

This command securely copied the file to the 'remote' system using encrypted SSH communication.

Figure 10: Test file securely transferred via SCP confirming file transfer capability.

Conclusion SSH

SSH is a secure protocol used for remote system administration and encrypted communication between computers. During this lab, SSH was installed, configured, and tested using a Kali Linux virtual machine. A secure connection was established, remote commands were executed, and SSH key authentication was implemented to enhance security. Additionally, secure file transfer was demonstrated using SCP. This exercise showed how SSH can be used for secure remote management and data transfer across networks

SSH Reflection

This lab gave me a practical understanding of how SSH addresses real-world security challenges. Before this lab, I understood SSH conceptually but had not fully appreciated how each feature maps to a specific security problem.

The first thing I learned was how SSH solves the problem of remote machine control in a safe and authenticated way. Being able to connect to localhost and execute commands remotely made this feel tangible rather than theoretical.

Setting up the RSA key pair was the most valuable part of the lab. It directly demonstrated why password-based authentication is a weakness. Passwords can be intercepted or brute-forced, whereas key-based authentication eliminates that risk entirely. Generating a 4096-bit key and using ssh-copy-id to deploy it showed me the full workflow a real system administrator would follow.

The SCP file transfer exercise reinforced that secure communication extends beyond just remote access. Moving files over an encrypted channel rather than an insecure protocol like FTP is a simple but critical security improvement in any professional environment.

Overall, this lab was effective at connecting theory to practice. Each step had a clear security purpose, and working through the setup manually gave me a much deeper understanding of how SSH protects systems than reading about it ever could.

Introduction to NMAP

Nmap is an open-source network scanning tool used to discover hosts, open ports, and running services on a network.

1. Check Nmap Version

The first step was verifying that Nmap was installed on the system. This was done using the command:

nmap --version

The output confirmed that Nmap was installed in the Kali Linux environment. Nmap is a network scanning tool used to discover hosts, services, and open ports on a network.

Figure 11: Nmap version 7.95 confirmed on Kali Linux

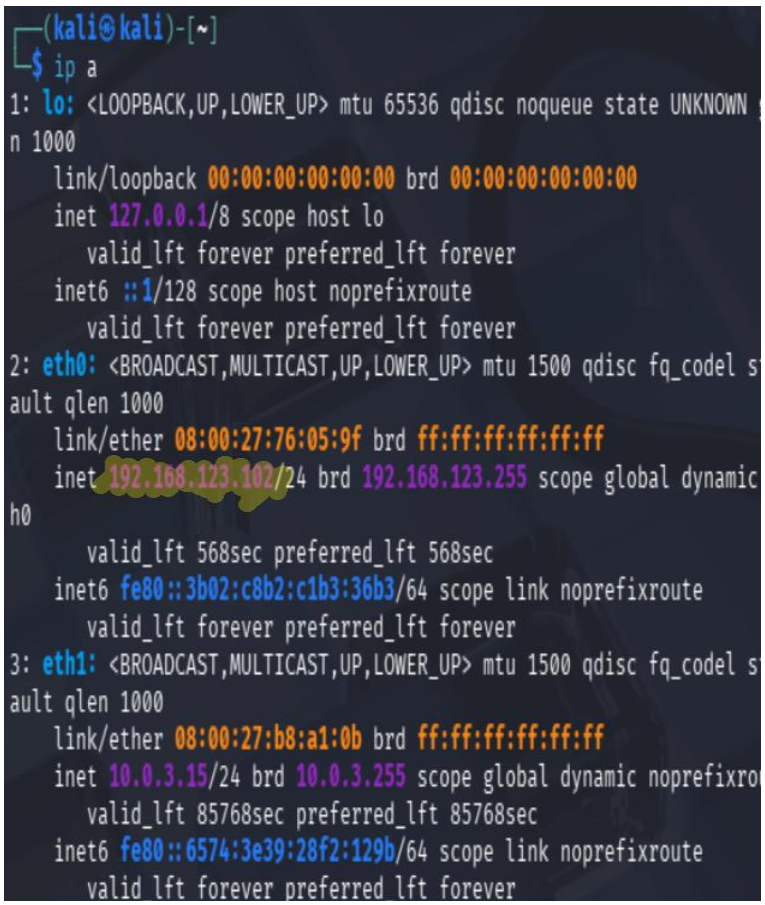


Figure 12: Kali Linux IP address identified as 192.168.123.102.

2. Find IP address

The IP address of the Kali Linux virtual machine was identified using the command:

ip a

The output showed that the system IP address was 192.168.123.102, indicating that the device is connected to the 192.168.123.0/24 network.

3. Host discovery

A host discovery scan was performed to identify active devices on the local network using the command:

nmap -sn 192.168.123.0/24

This scan checks all devices within the subnet and determines which hosts are active.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.123.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 10:59 EDT
Nmap scan report for 192.168.123.100
Host is up (0.00068s latency).
MAC Address: 08:00:27:D8:34:BF (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.123.102
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 7.32 seconds
```

Figure 13: Nmap ping sweep discovering 2 active hosts on the 192.168.123.0/24 network.

4. Basic Port Scanning

A basic port scan was performed using Nmap to identify open ports on the target system.

Command used:

nmap 192.168.123.102

The results showed that the host was active and that **port 22 was open**, which corresponds to the SSH service used for secure remote connections.

```
┌──(kali㉿kali)-[~]
└─$ nmap 192.168.123.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 11:01 EDT
Nmap scan report for 192.168.123.102
Host is up (0.0000010s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```

Figure 14: Default Nmap scan revealing port 22 open on the target.

5. Service Version Detection

Service detection was performed using the following command:

nmap -sV 192.168.123.102

This scan identified the specific version of the service running on the open port. The results showed that the system was running the **OpenSSH service**, which allows secure remote login.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.123.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 11:02 EDT
Nmap scan report for 192.168.123.102
Host is up (0.0000010s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.2p1 Debian 5 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/
submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.25 seconds
```

Figure 15: Service version scan identifying OpenSSH 10.2p1 running on port 22.

6. Operating System Detection

Operating system detection was performed using:

sudo nmap -O 192.168.123.102

The scan identified that the target system was running a Linux operating system. The network distance was reported as **0 hops**, indicating that the scan was performed on the local machine.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.123.102
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 11:04 EDT
Nmap scan report for 192.168.123.102
Host is up (0.000045s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submi
t/ .
Nmap done: 1 IP address (1 host up) scanned in 2.30 seconds
```

Figure 16: OS detection scan identifying the target as running Linux 5.0–6.2.

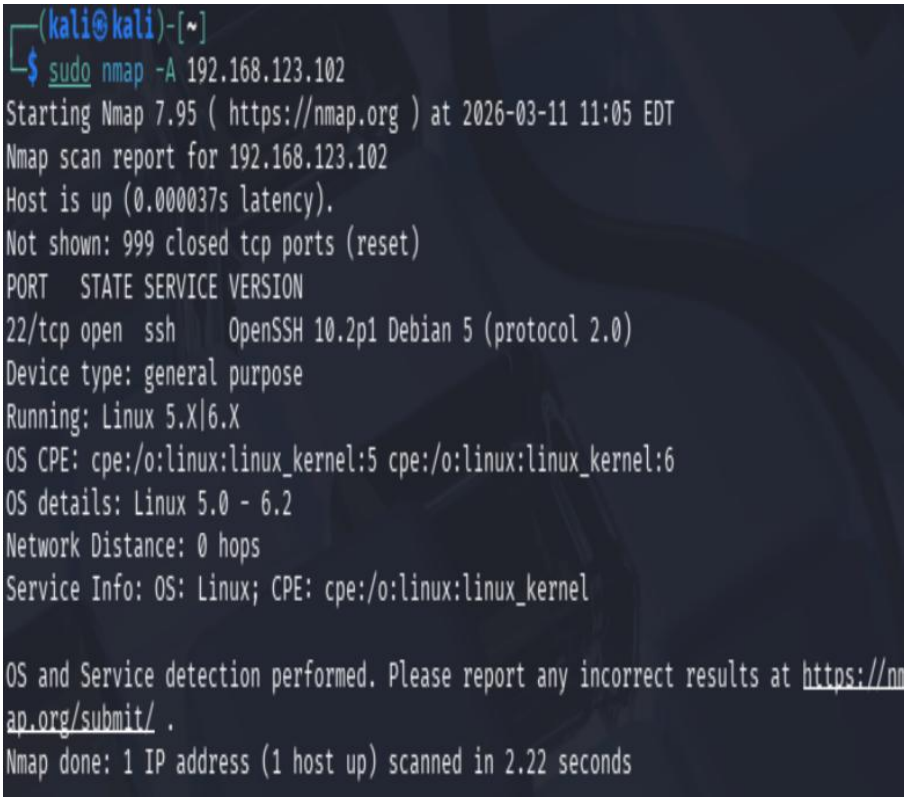
7. Aggressive Scan

An aggressive scan was performed using:

sudo nmap -A 192.168.123.102

This scan combines multiple scanning techniques, including service detection, operating system detection, script scanning, and traceroute.

The results confirmed that port 22 was open and running the OpenSSH service. The system was also identified as running a Linux operating system..



```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -A 192.168.123.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-11 11:05 EDT
Nmap scan report for 192.168.123.102
Host is up (0.000037s latency).
Not shown: 999 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 10.2p1 Debian 5 (protocol 2.0)
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.0 - 6.2
Network Distance: 0 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nm
ap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.22 seconds
```

Figure 17: Aggressive scan (-A) combining OS detection, version detection and service info in a single result.

Conclusion NMAP

Nmap is a powerful network scanning tool used for network discovery and security auditing. During this lab, several Nmap scanning techniques were performed using Kali Linux. The scans successfully identified the target host, detected open ports, identified running services, and determined the operating system. The results showed that port 22 was open and running the OpenSSH service on a Linux system. These techniques demonstrate how Nmap can be used by security professionals to analyse networks and identify potential security risks.

NMAP Reflection

This lab gave me hands-on experience with Nmap and showed me how it answers the fundamental questions a security analyst needs to ask when assessing any network. Before running a single scan, you cannot make any security decisions because you do not know what you are working with. Nmap solved this immediately, the host discovery scan revealed exactly which devices were active on the network, turning an unknown environment into a mapped one.

What struck me most was how quickly open ports were identified. In a real-world scenario, every open port is a potential entry point and seeing them listed in the scanned output made the concept of attack surface feel very concrete. A port being open does not necessarily mean it is vulnerable, but it does mean it demands attention.

Overall Nmap changed how I think about network security. Rather than seeing a network as a black box, I now see it as a collection of devices, ports and services, each of which can be systematically interrogated. That shift in perspective is exactly what this lab was designed to produce.

Use of Artificial Intelligence

In accordance with the University of Roehampton's AI guidelines, the following describes how AI tools were used to support the completion of this assessment.

ChatGPT was used for the following tasks:

- Reviewing and improving the overall organization and flow of the report to ensure it meets the assessment brief requirements.
- Debugging assistance during the lab environment setup, providing guidance on configuring the virtual machine network adapters and troubleshooting connectivity issues between the Kali Linux and WordPress VMs.

All technical work, screenshots, findings, and analysis represent my own independent lab work. AI was not used to generate technical findings, perform scans, or interpret screenshots.